

Question Paper

Exam Date & Time: 31-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025/JAN 2026
SUBJECT: CSE 3123/CSE_3123 OPERATING SYSTEMS
(SPL: COMPUTER SCIENCE & ENGINEERING)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Multiple users are transferring money and checking balances at the same time. (5)
- Explain how processes and interprocess communication can ensure data consistency.
 - Describe how threads can improve system responsiveness.
 - Identify potential race conditions and suggest methods to avoid them.

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 1B) A web browser runs multiple tabs performing heavy computations. Why are threads preferred over creating separate processes for each tab, and how does this affect CPU and memory usage? (3)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 1C) While booting, the OS loads device drivers and initializes network connections. Explain why threads are useful here instead of sequential execution. (2)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 2A) Consider the following set of processes: (5)

Process	Arrival Time	Execution Time	Priority
P ₁	0.0	8	2
P ₂	1.0	4	1
P ₃	2.0	9	3
P ₄	3.0	5	2

Solve:

- Apply Preemptive Priority Scheduling to these processes.
- Draw the Gantt chart for the schedule.
- Calculate the Average Turnaround Time (TAT) and Average Waiting Time (WT).

Note: Lower number = higher priority

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 2B) Consider a system with 3 processes (P₀, P₁, P₂) and 3 resource types (A, B, C). The system state is given below: (3)

Process	Max Need (A,B,C)	Allocation (A,B,C)
P ₀	(7, 5, 3)	(0, 1, 0)
P ₁	(3, 2, 2)	(2, 0, 0)
P ₂	(9, 0, 2)	(3, 0, 2)

The Available resources are: (A, B, C) = (3, 3, 2)

Solve:

- Determine whether the system is in a safe state using the Banker's Algorithm.
- If safe, provide a safe sequence of process execution.
- If P₁ requests (1, 0, 2) additional resources, determine whether the request can be granted immediately.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 2C) In a producer-consumer scenario, the buffer has a capacity of 5. Show how binary and counting semaphores can be used to prevent the producer from overwriting and the consumer from reading empty slots. (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3A) i) Explain with a neat diagram the operation of an inverted page table. Discuss the drawbacks of the conventional paging scheme for which the inverted page table can be used as a solution. (5)
ii) Consider the following segment table

Segment	Base	Length
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1952	96

What are the physical addresses for the following logical addresses?

- (a) 0, 430 (b) 1, 10 (c) 2, 500 (d) 3, 400
(3+2 = 5 marks)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 3B) Explain the following: (3)
(i) Second Chance page replacement algorithm
(ii) Enhanced Second Chance algorithm

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 3C) Describe the copy-on-write feature, and explain under what circumstances is it beneficial? (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4A) (i) Explain the following implementation methods for LRU page replacement algorithm (5)
(a) Counters (b) Stack
(ii) Consider the following page reference string:
7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1.
Illustrate the working of the following page replacement algorithms. How many page faults would occur for each of the replacement algorithms, with three page frames? Remember that all frames are initially empty, so your first unique pages will cost one fault each. Show the contents of page frames at each step.
(a) LRU replacement
(b) FIFO algorithm
(2+3 = 5 marks)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4B) Explain the need for disk scheduling. Given the following disk queue: 82, 170, 43, 140, 24, 16, 190 (3)
and the head initially at 50, compute the total head movement using FCFS and SSTF disk
scheduling algorithms.

Course Outcome: "CO5" / Cognitive Level: "Application"

- 4C) Explain the possible **locations of swap space** in Linux systems. With an example, justify how the **choice of location** affects the performance in different environments. (2)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5A) Illustrate how file-system mounting integrates secondary storage devices with an example. Analyze (5)
the possible security issues during mounting and its preventive measures.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- 5B) In a system implementing the Access Matrix model, three domains (D1, D2, D3) and three objects (3)
(File F1, File F2, and Printer P1) exist.
The following rights are defined:
• D1 can read F1 and write F2.
• D2 can read F1, copy that right to other domains, and print using P1.
• D3 can read and write F2 only when temporarily switched from D1 (domain switching is allowed).
Create an access matrix with given rights, and explain how it enforces a protection model.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- 5C) Define the principle of least privilege and justify its importance in protection design. (2)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-
2026

CSE_3123:Operating Systems

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	2
3	3	CO 3	8
4	4	CO 4	15
5	5	CO 5	15
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-
2026

CSE_3123:Operating Systems

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	3	10(20%)
3	Application	5	11(22%)
4	Analysis	7	29(58%)
5	Create	0	0(0%)

6	Evaluate	0	0(0%)
	Total	15	50 (100%)